— TECHNICAL DATA SHEET

NN-PyroShield STC-II

Non-Metal-Oxide (Organic) Intumescent Fire-Retardant Coating

Halogen-free organic APP / pentaerythritol / melamine intumescent system · water-based

NN-PyroShield STC-II — thick intumescent coating paste, supplied in pails

Document No.	Revision	Issue Date	Product Code
NNPL/TDS/PROD-08B	1.0	2 July 2026	STC-II

01 PRODUCT DESCRIPTION

NN-PyroShield STC-II is a non-metal-oxide (organic) intumescent fire-retardant coating for passive fire protection of structural steel, timber and other substrates. At normal temperature it forms a durable decorative-quality film; when exposed to fire (from ~200 °C) it reacts and swells many times its original thickness to form an insulating carbonaceous char that delays the temperature rise of the protected element and its structural failure.

Composition: Ammonium polyphosphate (acid source) + pentaerythritol (carbon source) + melamine (blowing agent) in a polymer binder — no metal-oxide additives, in a styrene-acrylic aqueous dispersion. A purely organic (non-metal-oxide) intumescent triad: the acid source, carbon source and blowing agent react on heating to form an expanded carbonaceous char, with low smoke and no metal-oxide fillers.

02 USES & APPLICATIONS

- **Structural steel —** Passive fire protection of structural steel columns, beams and hollow sections.
- **Timber —** Fire-retardant coating for timber, boards and other combustible building elements.
- **Fire rating —** Achieving specified fire-resistance ratings (30 / 60 / 90 / 120 min) as part of a certified system.
- **Construction —** Commercial, industrial and infrastructure projects requiring passive fire protection.

03 COMPOSITION (INTUMESCENT SYSTEM)

Component	Description
Acid source	Ammonium polyphosphate (APP)
Carbon source	Pentaerythritol (PER)
Blowing agent	Melamine
Binder	Styrene-acrylic aqueous dispersion
Char promoter	Organic char promoter (no metal oxides)

04 TECHNICAL PROPERTIES

Typical values for a water-based intumescent coating. Fire performance is DFT- and section-dependent; refer to the certified loading tables.

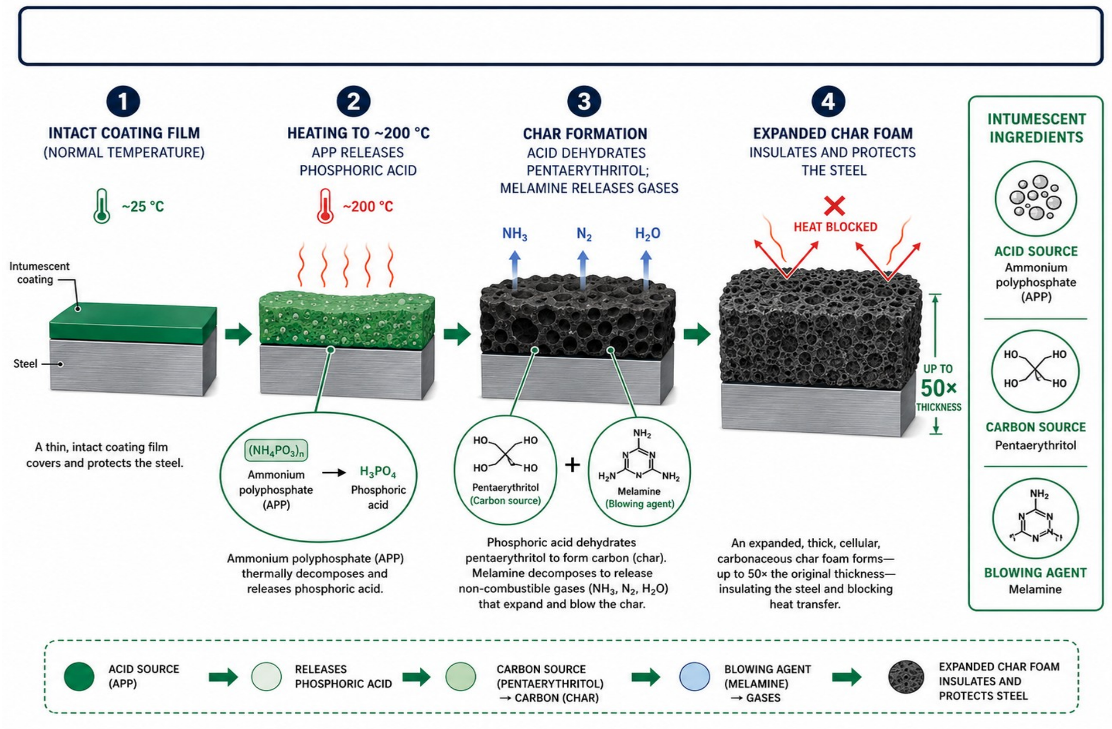
Property	Value
Appearance / finish	Off-white matt film (over-coatable)
Base / type	Water-based intumescent
Volume solids	≈ 65 – 72 %
VOC content	≤ 100 g/l (water-based)
Expansion (intumescence)	Up to ~50× original film thickness
Activation temperature	≈ 200 °C (onset of intumescence)
Fire-resistance ratings	30 / 60 / 90 / 120 minutes (system & DFT dependent)
Test standards	BS 476 Part 20/21 · IS 12777 (as applicable)
Dry film thickness (DFT)	≈ 0.4 – 3.0 mm (per section factor & rating)
Theoretical coverage	Per required DFT (from loading table)

05 APPLICATION

Intumescent coating applied to structural steel as part of a primer / intumescent / topcoat system

- **Preparation —** Surface prep: steel to be clean, dry and free from rust, mill-scale, oil and dust; blast-clean and apply a compatible anti-corrosive primer.
- **Application —** Apply by airless spray, brush or roller in coats to build the specified DFT; observe recoat and drying intervals.
- **Thickness control —** Measure wet-film thickness during application and verify dry-film thickness after cure with a calibrated gauge.
- **Topcoat —** Apply a compatible sealer / topcoat for humidity, weathering and decorative finish where required.
- **Conditions —** Do not apply below the specified minimum temperature or above the maximum humidity; avoid application on damp steel.

06 MECHANISM



Intumescent reaction: acid source + carbon source + blowing agent → expanded insulating char

On heating, the acid source (APP) decomposes to release phosphoric acid, which esterifies and dehydrates the carbon source (pentaerythritol) to form a carbon char. Simultaneously the blowing agent (melamine) releases inert gases that expand the softening char into a thick, cellular insulating foam. A purely organic (non-metal-oxide) intumescent triad: the acid source, carbon source and blowing agent react on heating to form an expanded carbonaceous char, with low smoke and no metal-oxide fillers.

07 PACKAGING, STORAGE & SHELF LIFE

Item	Detail
Packaging	20 / 25 kg pails (to order)
Storage	Store in a cool, dry, covered place between +5 °C and +40 °C; keep containers sealed; protect from frost
Shelf life	12 months in unopened, correctly stored containers
Handling	Stir before use; do not thin beyond the recommendation; protect from freezing

08 PRECAUTIONS

- **Handling —** Water-based and low-VOC; still avoid skin/eye contact and inhalation of spray mist — wear gloves, goggles and a mask.
- **Ventilation —** Ensure adequate ventilation during spray application; keep away from children and foodstuffs.

Refer to the NN-PyroShield STC-II Safety Data Sheet for full physical, ecological, toxicological and handling information.

Disclaimer: *The information and recommendations herein are given in good faith based on current knowledge. Fire performance depends on correct system build-up, dry film thickness and application; users must follow the certified loading tables and standards for their specific project. No warranty of merchantability or fitness for a particular purpose is implied. Verify suitability for the intended application and comply with all applicable fire-safety regulations.*